MultiMoE: Multimodal Mixture of Experts Adapter for Semantic Segmentation of Remote Sensing Images

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***Abstract*—Multimodal remote sensing image semantic segmentation has benefited remarkably from recent progress in deep learning. However, most existing methods treat heterogeneous modalities with excessive uniformity, failing to fully exploit their complementary features. Furthermore, the absence of visual prior knowledge limits their generalization capabilities across diverse scenarios. To address these challenges, we propose a novel multimodal fine-tuning framework based on vision foundation models, namely multimodal mixture-of-expert (MultiMoE). Specifically, we select the vision foundation model (DINOv3) as the frozen encoder and develop lightweight mixture-of-experts (MoEs) adapters on the encoder to transfer general visual knowledge to task-specific semantic contexts. These MoE-based adapters dynamically route samples to dedicated expert pools, thereby processing heterogeneous modalities distinctly and promoting complementary feature representation. Extensive experiments on two widely used benchmark datasets (ISPRS Vaihingen and Potsdam) demonstrate the competitive performance of our MultiMoE compared to state-of-the-art methods.**

***Index Terms*—Foundation model, mixture of experts (MoEs), multimodal adapter, semantic segmentation.**

I. Introduction

MULTIMODAL remote sensing image semantic segmentation aims to assign a semantic category to each image pixel by fusing contextual information extracted from various data modalities [1]. By integrating these complementary characteristics [2], multimodal semantic segmentation can achieve more accurate and reliable results compared to the single-modal pattern, providing essential support for applications such as land use and land cover classification [3], precision agriculture [4], environmental monitoring [5], and disaster management [6].

In recent years, deep learning has greatly advanced multimodal remote sensing image semantic segmentation methods. Initially, convolutional neural networks (CNNs) were dominant in this field due to their effectiveness in automatic feature extraction within encoder–decoder frameworks. These CNN-based methods typically use multiple independent branches to process inputs from different modalities, and then fuse the extracted semantic features to improve segmentation performance. With the advent of transformers [7], numerous transformer-based and CNN-transformer hybrid networks [8], [9] have been widely developed for multimodal remote sensing image semantic segmentation. Taking advantage of the unique self-attention mechanism, these approaches enhance the ability to capture global relationships between different data modalities and achieve superior multimodal feature fusion performance. However, the above CNNs and transformers were trained solely on task-specific datasets. This training scheme limits their acquisition of general visual knowledge and their adaptation to varying remote sensing scenes.

Fortunately, vision foundation models (VFMs) provide a new scheme for this challenge. Compared to CNNs and transformers, the VFMs demonstrate strong generalization ability due to rich semantic priors learned from large-scale and diverse pretraining. Driven by this, fine-tuning with foundation models emerges as a popular scheme for various segmentation tasks. For example, SegDINO [10] applies a frozen DINOv3 as the encoder and fine-tunes it with a trainable lightweight decoder for boosting natural and medical image semantic segmentation. Similarly, several SAM-based adaptation methods have also been developed for single-modality segmentation. For instance, CWSAM [11] and MeSAM [12] adapt SAM for remote sensing segmentation by customizing the image encoder and mask decoder.

The aforementioned fine-tuning methods, although obtaining remarkable performance in single-modality tasks, remain challenging on multimodal semantic segmentation. Existing multimodal fine-tuning approaches usually use independent Siamese branches or shared parameters to encode different modalities. Although this strategy can extract certain multimodal semantic features, it mostly treats different modalities in a nearly uniform manner and lacks cross-modal feature interaction in the encoding stage. Furthermore, heterogeneous

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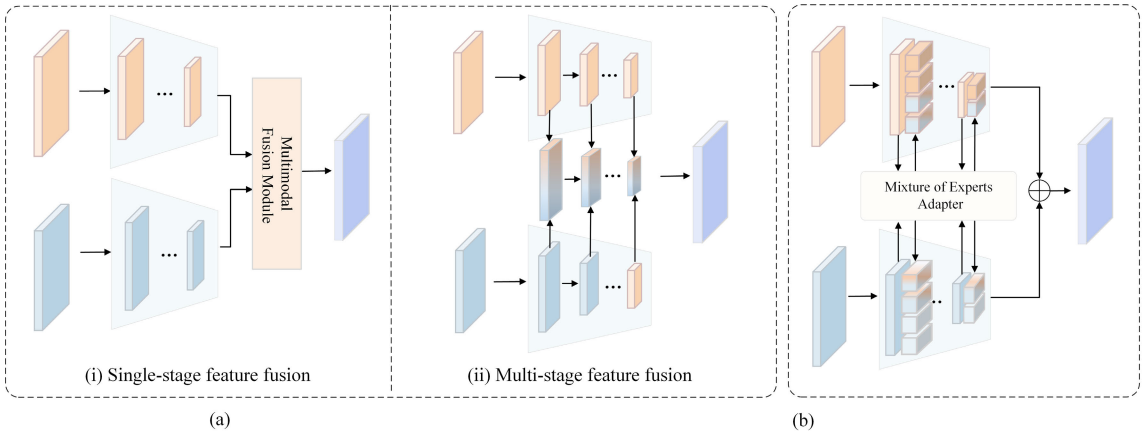


Fig. 1. Difference between conventional multimodal fusion frameworks and our multimodal interaction framework. (a) Traditional multimodal fusion frameworks usually aggregate features through single-stage or multistage fusion modules in a unidirectional manner. (i) Single-stage feature fusion. (ii) Multistage feature fusion. (b) In contrast, our framework replaces the multimodal fusion operation with adaptive multimodal feature interaction, introducing MoE-based adapters to enhance cross-modal feature interaction in the encoding stage.

modalities possess distinct imaging properties and structural characteristics. For instance, optical imagery often contains richer spectral and texture details, whereas DSM provides more informative geometric and elevation cues. Consequently, multimodal fine-tuning not only needs to extract modality-specific information, but also focuses on developing an interactive encoding mechanism to process heterogeneous modalities distinctly and achieve complementary feature representation.

Mixture-of-experts (MoEs) offer a natural solution to these challenges. Compared to the current Siamese structure, expert layers in MoE can process heterogeneous modalities distinctly through its dynamic parameter allocation mechanism [13]. This unique design allows MoE to model modality discrepancies and support complementary feature interaction. Following this idea, we propose a novel MoE-based fine-tuning framework, namely multimodal MoE (MultiMoE), for multimodal semantic segmentation of remote sensing images. Fig. 1 illustrates the difference between current multimodal fine-tuning methods and our MultiMoE. In conventional fusion strategies, different modalities are usually fused during decoding. This preserves modality-specific characteristics, but limits the multimodal interaction and the exploitation of complementary information. By contrast, MultiMoE encourages multimodal interaction within the encoding stage. Specifically, we select a vision foundation model (DINOv3) as the frozen encoder and develop MoE-based adapters on the encoder to transfer general visual knowledge to task-specific semantic contexts. We design a shared gating network with modality-specific MoE adapters to dynamically route samples to dedicated expert pools, thereby facilitating adaptive fine-tuning while preserving generalization capabilities.

The main contributions of this work are summarized as follows.

1) We develop a novel multimodal fine-tuning framework based on vision foundation models, namely MultiMoE, for multimodal semantic segmentation of remote sensing images. The proposed framework applies adaptive feature interaction within the frozen encoder, thereby achieving superior multimodal feature learning performance.
2) We propose a lightweight MoE-based adapter that performs dynamic expert routing to distinctly process heterogeneous modalities to achieve complementary feature representation.
3) Extensive experiments on two benchmark datasets, Vaihingen and Potsdam, verify the effectiveness of the proposed method and demonstrate competitive or superior segmentation performance compared to existing approaches.

The remainder of this article is organized as follows. Section II reviews the related work. Section III introduces the proposed MultiMoE framework and details its key components. Section IV presents extensive comparative experiments and ablation studies to evaluate the effectiveness of MultiMoE. Section V presents the model scale analysis. Finally, Section VI concludes the article.

II. Related Work

A. Multimodal Remote Sensing Image Semantic Segmentation

Semantic segmentation methods have developed rapidly, with most approaches relying on single-source remote sensing images. However, networks trained solely on feature representations derived from single-modal data often fail to meet the demands of contemporary remote sensing interpretation tasks [14], [15], making multimodal approaches increasingly popular.

Existing efforts in this area explored various architectures, including CNNs, transformer-based architectures, and hybrid architectures integrating CNNs and transformers [8], [9], [16], [17]. In multimodal tasks, different modalities are often processed with the encoder–decoder architectures [18], [19]. Early CNN-based methods, such as ResUNet-a [20], adopted a straightforward strategy by concatenating multimodal data at the input stage. Furthermore, FuseNet [16]

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introduced a dual-branch encoder architecture to incorporate depth features into an RGB network, significantly improving the accuracy of multimodal scene understanding. With the success of transformers, recent works have further explored transformer-based and hybrid architectures [21]. For instance, CMFNet [8] employed CNNs for feature extraction and used a cross-attention mechanism to fuse multimodal information at multiple scales. FTransUNet [17] integrated CNNs and ViTs to fuse shallow and deep features for multimodal representation learning.

Overall, despite the architectural diversity, these approaches mainly rely on similar and spatially uniform fusion strategies, treating multimodal features in a homogeneous manner. Such designs fail to capture the heterogeneous and content-dependent relationships across modalities, where the relative importance of each modality varies significantly across different spatial regions.

B. VFM in Remote Sensing

VFMs have recently achieved remarkable progress and demonstrated strong generalization ability across diverse visual tasks. By learning from large-scale data, VFMs capture general visual patterns and prior knowledge, enabling effective transfer to downstream applications [22]. Vision–language models such as CLIP [23], ALIGN [24], and SigLIP [25] learn aligned multimodal representations from massive image–text pairs. They have shown promising transferability to remote sensing tasks. In remote sensing, models such as SkySense [3] and ChangeCLIP [26] further demonstrate the potential of vision–language priors.

In the field of semantic segmentation, models such as SAM [27] introduce a promptable paradigm that enables flexible and general-purpose segmentation. The strong generalization capability of SAM has inspired a series of adaptations in the remote sensing domain, such as RingMo-SAM [28], MFNet [29]. These studies showed the potential of SAM for remote sensing image semantic segmentation, although its reliance on prompts and large model scale may limit efficiency in dense prediction tasks.

Beyond prompt-based models, self-supervised learning has emerged as another important paradigm for building VFMs without heavy annotation. Among various self-supervised foundation models, the DINO family has demonstrated powerful transferability across a wide range of visual tasks. DINO [30] and DINOv2 [31] have been extensively used for visual representation learning [32], owing to their ability to provide robust multiscale features that are well suited for downstream tasks such as detection [33] and segmentation [34]. DINOv3 [35] further improves representation quality and dense prediction ability, and is increasingly regarded as a powerful backbone for segmentation tasks. Recent work, such as Earth-adapter [36], further shows that parameter-efficient adaptation on top of DINO-based backbones can effectively tailor VFMs to remote sensing.

Nevertheless, effectively adapting such foundation models to remote sensing image semantic segmentation remains challenging due to domain gaps, complex spatial patterns, and the heterogeneity among multimodal data. How to fully exploit the rich representations of models like DINOv3 while incorporating remote sensing characteristics is still an open research problem.

C. Mixture of Experts

MoEs consists of a set of expert networks and a gating network [37]. The gating network selects a sparse weighted combination of experts. Benefiting from the sparse architecture of MoE, recent methods [13], [38], [39] have improved model capacity while maintaining computational efficiency.

In remote sensing, MoE has been applied to address data heterogeneity and domain variability. For example, GRAM [40] utilized region-aware experts to capture region-specific characteristics while maintaining shared representations for generalization. Earth-adapter [36] introduced a mixture-of-adapters design to handle remote sensing artifacts by decomposing features into different subspaces and adaptively combining expert outputs.

However, existing methods mainly focused on region-level or feature-level specialization and did not explicitly consider multimodal discrepancies. These limitations motivate the design of a MultiMoE adapter with a shared gating network, in which experts can specialize in different modalities to better exploit complementary information.

III. Methodology

The overall architecture of the proposed MultiMoE is illustrated in Fig. 2. The multimodal inputs to MultiMoE are separately processed by the image encoder, which is equipped with a MultiMoE adapter inserted between selected transformer blocks. The MoE adapter enhances modality-specific representations through dynamic expert routing while maintaining shared gating across modalities, thereby effectively leveraging complementary multimodal information for semantic segmentation. The resulting features are fed into a lightweight decoder [41], which reconstructs the high-level semantic information into the final segmentation map by jointly modeling global and local contexts.

A. Encoder

As illustrated in Fig. 2, we apply a frozen DINOv3 [35] as the encoder for processing both RGB and DSM modalities. Given RGB images $X \in \mathbb{R}^{H\times W\times 3}$ and the corresponding DSM depth data $Y \in \mathbb{R}^{H\times W\times 1}$, where H and W represent the height and width of the inputs, respectively, each modality is independently processed by the encoder.

The encoder adopts a nonhierarchical ViT backbone, which maintains a constant spatial resolution for image embeddings. The input is first divided into $(H \times W)/P^2$ patches ($P = 16$), and each patch is projected into a d-dimensional embedding. Subsequently, the resulting token sequence is processed by L (default = 24) stacked transformer blocks, denoted by ViT-s_l, where $l \in \{1, \ldots, L\}$. It is worth noting that the output features of each DINOv3-ViT block are adaptively processed by the MoE adapter, and the resulting features are then fed into the next ViT block. This mechanism facilitates coordinated processing between RGB representations with rich textures

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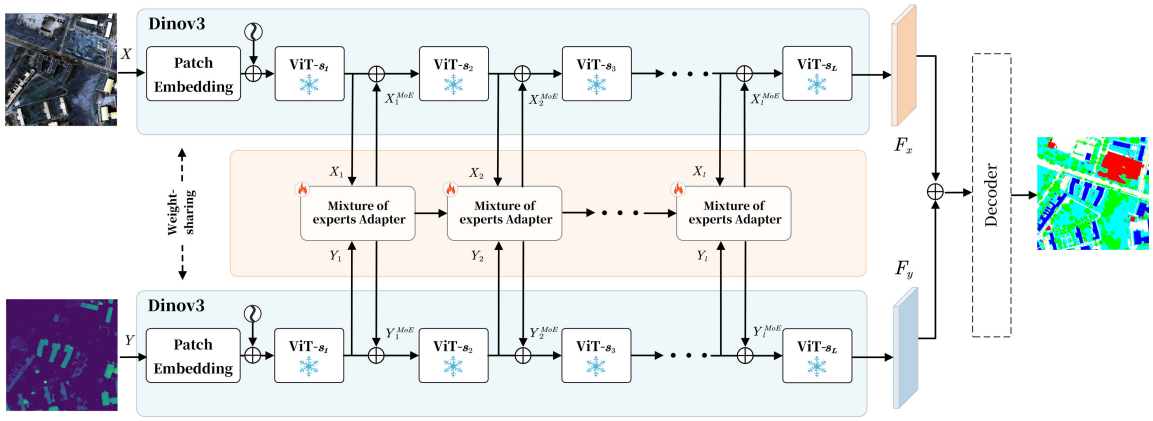


Fig. 2. Overview of the proposed MultiMoE framework, which consists of a frozen DINOv3 image encoder, MoE adapter modules composed of shared gating and modality-specific experts, and a general decoder. During training, the backbone remains frozen while the MoE adapters and decoder are learnable.

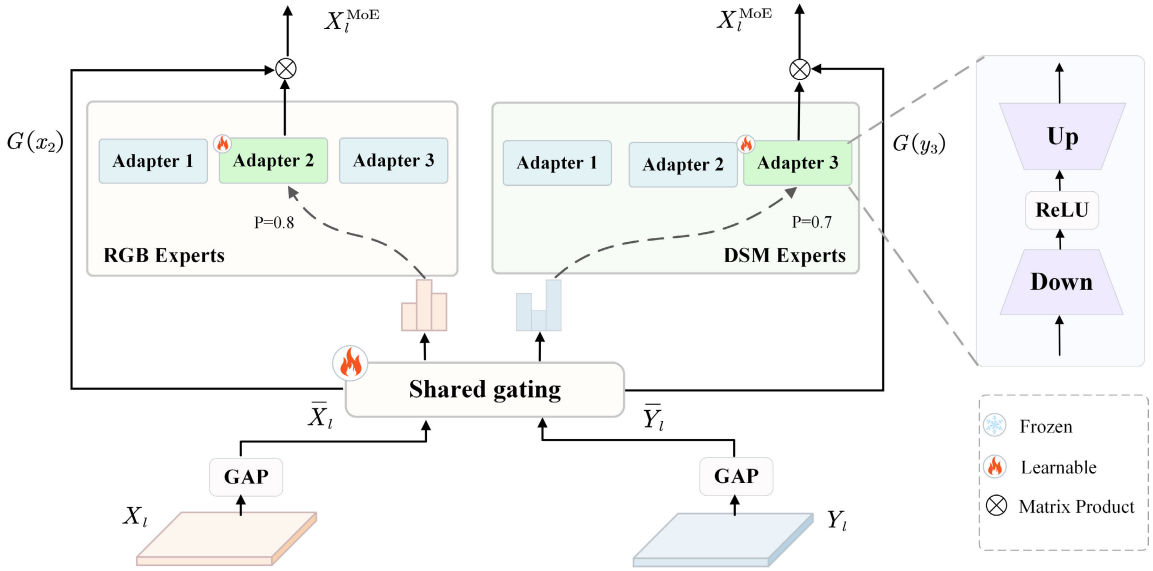


Fig. 3. Illustration of the MoE adapter. We diagram two modality features (X_l for RGB, Y_l for DSM) extracted from the lth ViT block. They are first globally average pooled and then routed across the corresponding expert pools through the shared gating network. The MultiMoE adapter returns the output of the selected expert multiplied by the corresponding gate value.

and DSM features with complementary spatial information. The shared gating MoE adapter enables consistent expert routing across modalities, preserving modality-specific characteristics for improved multimodal representation learning. The multimodal features at the output of the lth DINOv3-ViT block are denoted as $X_l, Y_l \in \mathbb{R}^{H/P \times W/P \times d}$.

B. MultiMoEs Adapters

We leverage MoE to build an architecture that extends DINOv3 to multimodal segmentation and enables effective cross-modal interaction through a shared gating function. Our MoE adapters are inserted after every transformer block. As shown in Fig. 3, the proposed module is composed of separate adapter expert pools for each modality and a shared gating network. Completely independent MoE architectures would isolate the modalities, thereby discarding crucial multimodal interaction. Conversely, forcing heterogeneous modalities to share a single MoE or experts often leads to feature conflicts and training instability due to their distinct physical properties. To resolve this problem, our design employs a shared gating network coupled with modality-specific expert pools.

1) Adapters as Experts: MoE in vision-language models usually incorporate the experts inside networks, which can be FFNs or attention heads [37], [38], [42]. However, replacing a dense layer in DINOv3 may incur high computational cost due to the full-parameter tuning. Prior works [43], [44] demonstrate that light adapter modules can significantly improve the adaptation efficiency of vision–language models for downstream tasks. Inspired by this, we employ the adapter, which consists of a down-projection, a ReLU activation, and an up-projection, as the expert in the MoE to enhance multimodal learning with DINOv3. The feature adaptation performed by the eth expert can be written as

$$E_e(h) = \mathrm{ReLU}(\mathrm{LN}(h) W_d) W_u \tag{1}$$

where $W_d \in \mathbb{R}^{c \times \hat{c}}$ and $W_u \in \mathbb{R}^{\hat{c} \times c}$ are the down-projection and up-projection matrices, respectively. For multimodal inputs,

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we construct two modality-specific expert pools $\{E_e^x\}_{e=1}^E$ and $\{E_e^y\}_{e=1}^E$ for RGB and DSM, respectively, with identical structures but independent parameters.

2) Shared Gating and Modality-Specific Experts: The MoE adapter consists of a shared gating network G and two sets of modality-specific expert networks, denoted by $\{E_1^x, \ldots, E_E^x\}$ and $\{E_1^y, \ldots, E_E^y\}$ for RGB and DSM, respectively. Let X_l and Y_l denote the RGB and DSM features at the lth DINOv3 block. We first obtain their global representations $\bar{X}_l$ and $\bar{Y}_l$ via average pooling.

In the MoE adapter, each sample is routed to activate the Top-k experts out of E available experts. The routing decisions are performed at the sample level using globally pooled representations, and all tokens within the same modality share the identical selected experts. The routing weights are produced by a shared gating network

$$G\left(\bar{X}_l\right) = \text{Softmax}\left(W_g \bar{X}_l\right) \tag{2}$$

$$G\left(\bar{Y}_l\right) = \text{Softmax}\left(W_g \bar{Y}_l\right) \tag{3}$$

where the trainable weight matrix $W_g \in \mathbb{R}^{D\times E}$ is shared across modalities. Based on the routing scores, the MoE outputs are computed as weighted combinations of the selected experts

$$X_l^{\text{MoE}} = \sum_{e=1}^{E} G\left(\bar{X}_l\right)_e \cdot E_e^x\left(X_l\right) \tag{4}$$

$$Y_l^{\text{MoE}} = \sum_{e=1}^{E} G\left(\bar{Y}_l\right)_e \cdot E_e^y\left(Y_l\right) \tag{5}$$

where $G(\cdot)_e$ denotes the gating weight for the eth expert, and only the Top-k experts have nonzero gating weights. Although the gating parameters W_g are shared, the routing decisions remain content-dependent due to the differences between RGB and DSM inputs. The MoE outputs are integrated with the original features via residual connections

$$X_l' = X_l + \alpha \cdot X_l^{\text{MoE}} \tag{6}$$

$$Y_l' = Y_l + \alpha \cdot Y_l^{\text{MoE}} \tag{7}$$

where α is a learnable scaling factor controlling the contribution of the MoE adapter.

3) Load Balancing Loss: To prevent the gating network from collapsing to a small subset of experts, we impose a soft constraint on the batch-wise average routing probabilities at each MoE layer. For the lth layer and modality m, let $\mathbf{p}_i^{(l,m)} \in \mathbb{R}^E$ denote the routing probability over E experts for the ith sample. We define the expert importance of a mini-batch as

$$\text{Imp}^{(l,m)} = \frac{1}{B}\sum_{i=1}^{B} \mathbf{p}_i^{(l,m)} \tag{8}$$

where $\text{Imp}^{(l,m)} \in \mathbb{R}^E$ is the batch-wise average routing probability over experts. The load balancing loss is then defined as

$$\mathcal{L}_{\text{lb}}^{(l,m)} = CV\left(\text{Imp}^{(l,m)}\right)^2 \tag{9}$$

where

$$CV\left(\text{Imp}^{(l,m)}\right) = \frac{\text{Std}\left(\text{Imp}^{(l,m)}\right)}{\text{Mean}\left(\text{Imp}^{(l,m)}\right) + \epsilon}. \tag{10}$$

For dual-modal inputs, the loss is computed independently for each modality and averaged within each layer. The final load balancing objective is obtained by averaging over all MoE layers

$$\mathcal{L}_{\text{lb}} = \frac{1}{L}\sum_{l=1}^{L}\frac{1}{M}\sum_{m=1}^{M}\mathcal{L}_{\text{lb}}^{(l,m)}. \tag{11}$$

Accordingly, we incorporate the load balancing loss into the training objective as an auxiliary regularization term. The overall loss $\mathcal{L}$ is a combination of the cross-entropy loss $\mathcal{L}_{\text{ce}}$ and the load balancing loss $\mathcal{L}_{\text{lb}}$

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \lambda_{\text{lb}}\mathcal{L}_{\text{lb}} \tag{12}$$

where λ_{lb} is set to 0.01 by default.

IV. Experiments

A. Experimental Datasets

1) Vaihingen: The Vaihingen dataset consists of 33 fine-resolution TOP orthophotos, each with an average size of 2500 × 2000 pixels. It includes three multispectral bands, namely near-infrared, red, and green (IRRG), along with a normalized DSM at a ground sampling distance (GSD) of 9 cm. The dataset contains five foreground classes (impervious surface, building, low vegetation, tree, and car) and one background class (clutter). We split the 16 patches with eroded boundary data provided by ISPRS into a training set of 12 patches and a test set of four patches. Instead of conventional cropping, a 256 × 256 sliding window was adopted to efficiently process large-scale patches, resulting in a processed dataset of approximately 960 training images and 320 test images.

2) Potsdam: The Potsdam dataset consists of 24 high-resolution orthophotos, each with a size of 6000 × 6000 pixels, and contains the same classes as the Vaihingen dataset. These orthophotos include four channels, namely red, green, blue, and near-infrared, as well as a normalized DSM with a GSD of 5 cm. We split the 24 patches into 18 patches for training and six patches for testing. Using the same sliding-window approach, this dataset comprises 10 368 training samples and 3456 test samples. In addition, a smaller stride was adopted when collecting training samples, and predictions over overlapping regions were averaged during testing to reduce boundary effects. Visual examples from both datasets are presented in Fig. 4.

B. Experimental Metrics

To assess the semantic segmentation results of the multimodal remote sensing data, we employed three evaluation metrics, including overall accuracy, mean $F1$ score, and mean intersection over union. These standard statistical indices allow for fair comparisons between the performance of the proposed MultiMoE and state-of-the-art baseline methods. In particular, we compute mF1 and mIoU of the five foreground classes, while OA evaluates both foreground classes and the background class. The evaluation metrics are defined as follows:

$$F1 = 2 \times \frac{precision \times recall}{precision + recall} \tag{13}$$

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TABLE I

QUANTITATIVE RESULTS ON THE VAIHINGEN DATASET. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE (%)

Method	Backbone	OA						mF1	mIoU
		Bui.	Tre.	Low.	Car.	Imp.	Total		
FuseNet [16]	VGG16	96.28	90.28	78.98	81.37	91.66	90.51	87.71	78.71
vFuseNet [45]	VGG16	95.92	91.36	77.64	76.06	91.85	90.49	87.89	78.92
MAResU-Net [46]	ResNet18	94.84	89.99	79.09	85.89	92.19	90.17	88.54	79.89
ESANet [47]	ResNet34	95.69	90.50	77.16	85.46	91.39	90.61	88.18	79.42
CMGFNet [48]	ResNet34	97.75	91.60	80.03	87.28	92.35	91.72	90.00	82.26
PSPNet [49]	ResNet101	94.52	90.17	78.84	79.22	92.03	89.94	86.55	76.96
SA-GATE [50]	ResNet101	94.84	92.56	81.29	87.79	91.69	91.10	89.81	81.27
CMFNet [8]	VGG16	97.17	90.82	80.37	85.47	92.36	91.40	89.48	81.44
UNetFormer [41]	ResNet18	96.23	91.85	79.95	86.99	91.85	91.17	89.48	81.97
MFTransNet [9]	ResNet34	96.41	91.48	80.09	86.52	92.11	91.22	89.62	81.61
TransUNet [51]	R50-ViT-B	96.48	92.77	76.14	69.56	91.66	90.96	87.34	78.26
FTransUNet [17]	R50-ViT-B	98.20	91.94	81.49	91.27	93.01	92.40	91.21	84.23
RS3Mamba [52]	R18-Mamba-T	97.40	92.14	79.56	88.15	92.19	91.64	90.34	82.78
FTransDeepLab [53]	ResNet101	98.11	93.45	80.35	89.98	93.23	92.61	91.00	83.87
MultiMoE (Ours)	Dinov3-ViT-L	95.39	93.74	81.67	94.21	93.48	92.16	91.41	84.47

TABLE II

QUANTITATIVE RESULTS ON THE POTSDAM DATASET. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE (%)

Method	Backbone	OA						mF1	mIoU
		Bui.	Tre.	Low.	Car.	Imp.	Total		
FuseNet [16]	VGG16	97.48	85.14	87.31	96.10	92.64	90.58	91.60	84.86
vFuseNet [45]	VGG16	97.23	84.29	89.03	95.49	91.62	90.22	91.26	84.26
MAResU-Net [46]	ResNet18	96.82	83.97	87.70	95.88	92.19	89.82	90.86	83.61
ESANet [47]	ResNet34	97.10	85.31	87.81	94.08	92.76	89.74	91.22	84.15
CMGFNet [48]	ResNet34	97.41	86.80	86.68	95.68	92.60	90.21	91.40	84.53
PSPNet [49]	ResNet101	97.03	83.13	85.67	88.81	90.91	88.67	88.92	80.36
SA-GATE [50]	ResNet101	96.54	81.18	85.35	96.63	90.77	87.91	90.26	82.53
CMFNet [8]	VGG16	97.63	87.40	88.00	95.68	92.84	91.16	92.10	85.63
UNetFormer [41]	ResNet18	97.69	86.47	87.93	95.91	92.27	90.65	91.71	85.05
MFTransNet [9]	ResNet34	97.37	85.71	86.92	96.05	92.45	89.96	91.11	84.04
TransUNet [51]	R50-ViT-B	96.63	82.65	89.98	93.17	91.93	90.01	90.97	83.74
FTransUNet [17]	R50-ViT-B	97.78	88.27	88.48	96.31	93.17	91.34	92.41	86.20
RS3Mamba [52]	R18-Mamba-T	97.70	86.11	89.53	96.23	91.36	90.49	91.69	85.01
FTransDeepLab [53]	ResNet101	97.58	85.87	90.08	96.94	92.81	90.97	92.08	85.62
MultiMoE (Ours)	Dinov3-ViT-L	98.66	88.61	91.45	96.30	93.36	92.39	93.32	87.74

$$IoU = \frac{TP}{TP + FP + FN} \tag{14}$$

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \tag{15}$$

where TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively. Furthermore, *precision* and *recall* are given by

$$precision = \frac{TP}{TP + FP} \tag{16}$$

$$recall = \frac{TP}{TP + FN}. \tag{17}$$

C. Implementation Details

All experiments were conducted using PyTorch on a single NVIDIA GeForce RTX 4090 GPU with 24 GB of RAM. The stochastic gradient descent (SGD) algorithm was used to train the model. The base learning rate was set to 0.01, with differentiated learning rates for different parameter groups: gate and residual scaling parameters were trained at 0.001 to prevent early routing overfitting, while expert and other parameters were trained at the base rate. A multistep learning rate scheduler was applied. All models were trained for a total of 50 epochs with a batch size of 10. Basic data augmentation techniques, including random rotation and flipping, are applied to image patches extracted with a sliding window of size 256 × 256 pixels. For the MoE variants, we employ a frozen pretrained vision transformer (ViT-L/16) backbone from DINOv3 for feature extraction. The MoE adapter is configured with E = 3 experts and a top-1 gating strategy. A dual-modality load balancing loss is introduced with a balance weight $\lambda_{lb} = 0.01$ to encourage diverse expert utilization across different modalities.

D. Quantitative Comparison

To comprehensively evaluate the performance of the proposed MultiMoE, we compare it with 14 state-of-the-art methods, including FuseNet [16], vFuseNet [45], MAResUNet [46], ESANet [47], CMGFNet [48], PSPNet [49],

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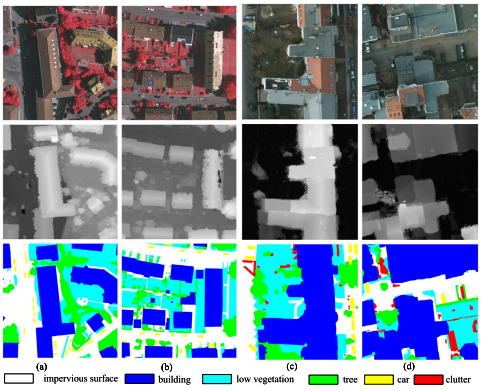


Fig. 4. Image samples of size 1024 × 1024 from (a) and (b) Vaihingen and (c) and (d) Potsdam, respectively. The first row shows the orthophotos with three channels (IRRG for Vaihingen and RGB for Potsdam). The second and third rows show the corresponding depth information and semantic labels pixel-wise. These samples present the auxiliary role of the DSM modality for classes with consistent elevation and clear boundaries.

SA-GATE [50], CMFNet [8], UNetFormer [41], MFTransNet [9], TransUNet [51], FTransUNet [17], RS^{3}Mamba [52], and FTransDeepLab [53]. In our experiments, PSPNet, MAResU-Net, UNetFormer, and RS^{3}Mamba only considered the optical image. This setting highlights the influence of DSM data and illustrates the advantages of multimodal over single-modal. By contrast, the DSM is selected as the second modality input in other approaches. The comparative quantitative results are presented in Tables I and II.

1) Comparisons on the Vaihingen Dataset: As presented in Table I, our MultiMoE yields a 91.41% mF1 and an 84.47% mIoU, outperforming all competing methods on both metrics. In particular, it surpasses the recent state-of-the-art method FTransUNet by 0.20% in mF1 and 0.24% in mIoU. Although the overall accuracy is not the highest, the superior mF1 and mIoU still demonstrate the effectiveness of MultiMoE. The relatively limited gains on the *building* class suggest that boundary and roof-structure details are not yet fully optimized on the Vaihingen dataset. Nevertheless, MultiMoE remains competitive on $Tree$, $Impervious\ Surface$, and *Low Vegetation*, with its clearest advantage on *Car*, where the 2.94% improvement demonstrates its ability to capture small-object classes.

Fig. 5 presents a visual comparison of the results produced by various methods and the proposed MultiMoE. Our MultiMoE demonstrates superior performance in segmenting impervious surfaces, producing more complete regions and clearer boundaries.

2) Comparisons on the Potsdam Dataset: The experimental results on the Potsdam dataset are presented in Table II. The proposed MultiMoE achieved OA of 92.39%, mF1 of 93.32%, and mIoU of 87.74%, which corresponds to increases of 1.05%, 0.91%, and 1.54% as compared to the FTransUNet, respectively. On the Potsdam dataset, MultiMoE improves the *Low vegetation* class by 1.37% compared with FTransDeepLab. The proposed MultiMoE leverages sparse, input-dependent routing to encourage emergent expert specialization, allowing them to focus on modality-specific characteristics.

Fig. 6 shows a visualization example from the Potsdam dataset, where clearer recognition of vegetation and impervious surfaces can be observed. These visual improvements are consistent with the significant gains in mIoU and other metrics reported in Table II.

E. Ablation Study

1) Effectiveness of Auxiliary Modality and MoE Adapter: To illustrate the necessity of multimodal learning, we first conducted ablation studies on the auxiliary modality. The results are presented in Table III. With the deployment of the auxiliary modality (i.e., DSM), we obtained an increase of 4.68% mIoU on Vaihingen and 3.02% on Potsdam, which demonstrates the effectiveness and necessity of the auxiliary modality.

To verify the effectiveness of the proposed MoE adapter, we compared three variants: the DINOv3 backbone, the backbone with a single adapter, and the backbone with the proposed MultiMoE adapter. As shown in Table IV, with the deployment of a lightweight single adapter, the Vaihingen mIoU increased by 0.27%, while the Potsdam mIoU decreased by 0.33%. This indicates that adding a simple adapter provides limited adaptation capability. Moreover, the employment of the MoE adapter can provide further improvement of 3.26% mIoU on Vaihingen and 1.99% on Potsdam compared to the DINOv3 baseline, which indicates that the proposed MoE adapter effectively enhances multimodal feature interaction and representation capability.

To further evaluate the effectiveness of MultiMoE, we visualize the feature distributions of the baseline model and MultiMoE on the Vaihingen dataset. Fig. 7 compares the feature responses of the baseline and MultiMoE on the Vaihingen dataset. The baseline produces scattered activations for both IRRG and DSM inputs. In contrast, MultiMoE yields more concentrated and spatially coherent responses. Furthermore, the DSM heatmaps place more emphasis on roads and geometric structures, indicating that DSM provides complementary information. Therefore, MultiMoE can better exploit cross-modal complementarity for remote sensing semantic segmentation.

2) Ablation Study on the Gating Network: To demonstrate the effectiveness of the shared gating network, we conduct an ablation study, comparing our method (w/SG) against both the independent gating network (w/IG) and the simple average baseline (w/o G). As illustrated in Table V, using this simple averaging strategy results in an mIoU drop of 2.75% on the Vaihingen dataset and 2.78% on the Potsdam dataset, which reveals that simply increasing the number of experts brings limited performance gains, while effective expert selection via dynamic routing is more critical. Although the independent gate yields improvements over the baseline, its performance remains inferior to that of the shared gating network.

In Fig. 8, the class-wise expert usage heatmaps demonstrate that different semantic classes and modalities exhibit highly distinct routing preferences. Meanwhile, the overall expert selection frequency across both datasets in Fig. 9 remains generally balanced, verifying the efficacy of our

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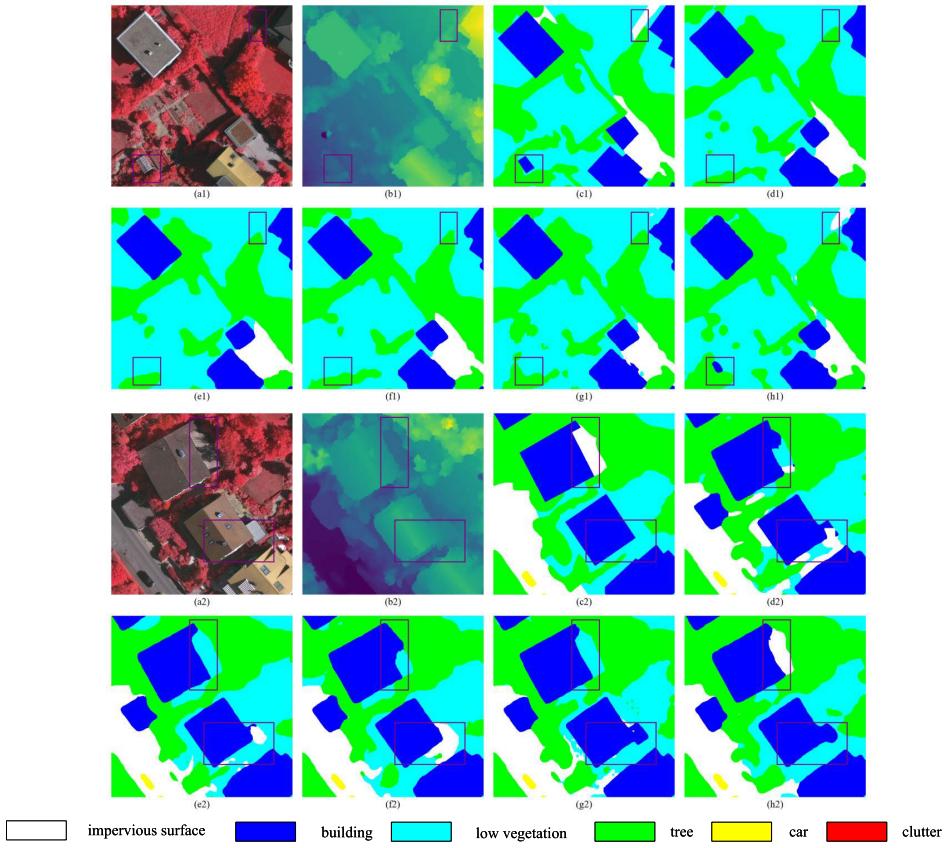


Fig. 5. Visualized comparison on the Vaihingen dataset with a size of 512 × 512. (a) IRRG images, (b) DSM, (c) ground truth, (d) FTransUNet, (e) MF-TransNet, (f) CMGFNet, (g) FTransDeepLab, and (h) proposed MultiMoE. To distinguish the key differences, purple boxes are added to all subfigures.

TABLE III

QUANTITATIVE RESULTS OF THE VAIHINGEN AND POTSDAM DATASETS WITH DIFFERENT MODALITIES WITH DINOV3 BASELINE (%)

Dataset	Bands	OA (%)						mF1	mIoU
		Bui.	Tre.	Low.	Car.	Imp.	Total		
Vaihingen	IRRG	94.99	90.43	78.26	70.40	91.68	90.00	86.24	76.53
	IRRG+DSM	95.43	92.68	79.73	88.03	90.49	90.74	89.42	81.21
Potsdam	RGB	97.88	87.40	86.62	92.17	90.31	90.31	90.38	82.73
	RGB+DSM	97.97	88.03	88.31	94.53	93.32	91.32	92.16	85.75

TABLE IV

COMPARISON BETWEEN THE DINOV3 BASELINE, A LIGHTWEIGHT SINGLE ADAPTER (SINGLE E), AND THE PROPOSED MULTIMOE ON THE VAIHINGEN AND POTSDAM DATASETS (%)

Dataset	Method	OA (%)						mF1	mIoU
		Bui.	Tre.	Low.	Car.	Imp.	Total		
Vaihingen	w/o MoE	95.43	92.68	79.73	88.03	90.49	90.74	89.42	81.21
	Single E	94.72	90.74	83.33	88.79	91.73	90.96	89.59	81.48
	w MoE	95.39	93.74	81.67	94.21	93.48	92.16	91.41	84.47
Potsdam	w/o MoE	97.97	88.03	88.31	94.53	93.32	91.32	92.16	85.75
	Single E	97.88	85.98	89.70	94.82	91.80	91.06	91.96	85.42
	w MoE	98.66	88.61	91.45	96.30	93.36	92.39	93.32	87.74

shared gating network. To investigate the importance of expert specialization, Fig. 10 visualizes the sample-level gating distribution on Vaihingen tile 5 (Layer 8). The distinct gating decisions between RGB and DSM inputs clearly reveal tailored experts for land-cover semantics and elevation-based structures, respectively.

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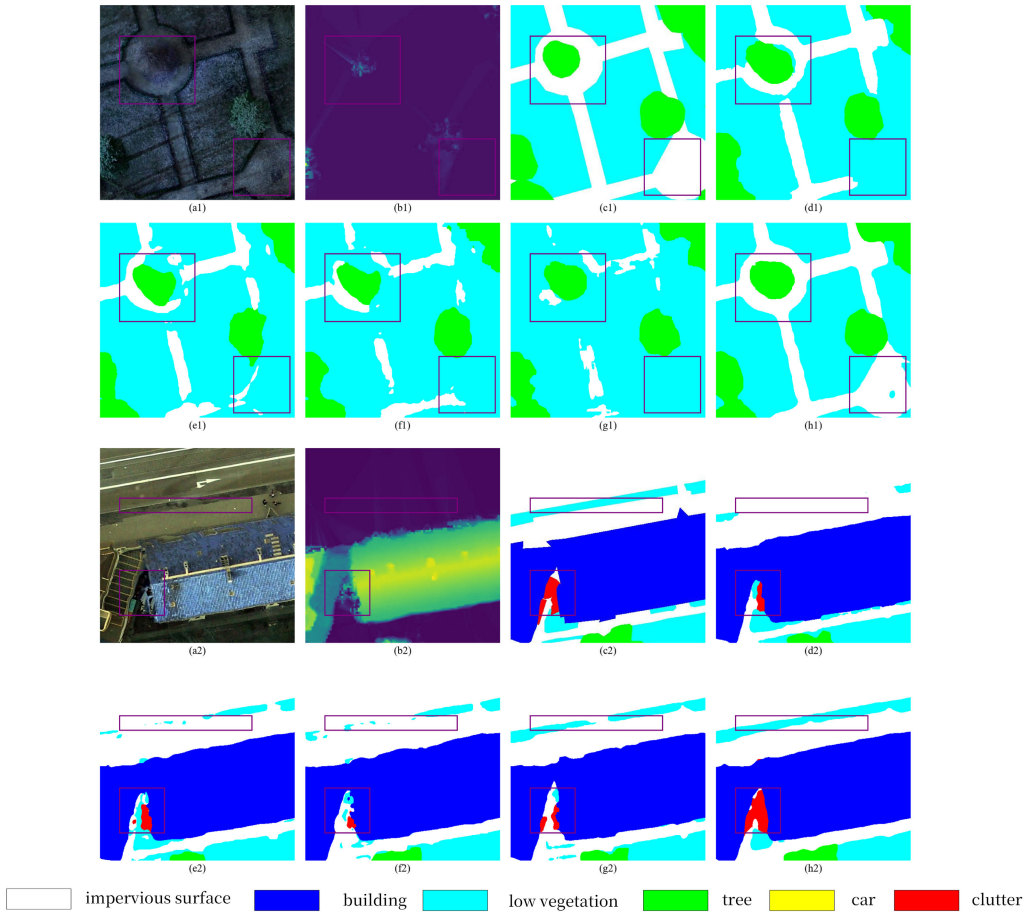


Fig. 6. Visualized comparison on the Potsdam dataset with the size of 512×512. (a) RGB images, (b) DSM, (c) ground truth, (d) FTransUNet, (e) MFTransNet, (f) CMGFNet, (g) FTransDeepLab, and (h) proposed MultiMoE. To distinguish the key differences, purple boxes are added to all subfigures.

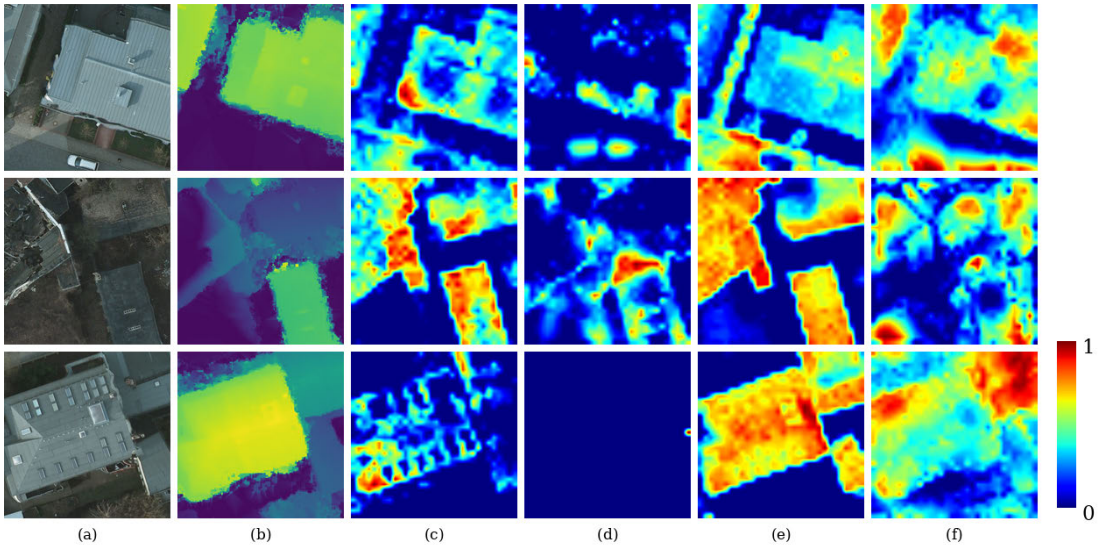


Fig. 7. Visualization of feature response heatmaps on the Potsdam dataset. (a) RGB images, (b) DSM; (c) heatmaps from RGB images, (d) heatmaps from DSM, both generated by the original Dinov3; (e) heatmaps from RGB images, and (f) heatmaps from DSM, both generated by the proposed MoE. High-value regions in the heatmaps correspond to areas identified as buildings. The effectiveness of our MultiMoE can be clearly observed.

3) Impact of Expert Number and Activated Experts on Performance: To investigate the impact of the number of experts (E) and the Top-k routing strategy, we perform experiments with different configurations. As shown in Table VI, Top-1 achieves the highest mIoU on both datasets. Specifically, with $E = 3$, Top-1 routing brings mIoU gains

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TABLE V

COMPARISON RESULTS UNDER DIFFERENT GATE CONFIGURATIONS, THE MODEL WITHOUT GATING NETWORK (W/O G), THE MODEL WITH SHARED GATING NETWORK (W/SG), AND WITH INDEPENDENT GATING NETWORK (W/IG) ON THE VAIHINGEN AND POTSDAM DATASETS (%)

Method	Vaihingen			Potsdam		
	OA	mF1	mIoU	OA	mF1	mIoU
w/o G	91.12	89.74	81.72	90.69	91.66	84.96
w / IG	92.52	91.04	83.92	92.33	93.20	87.53
w / SG	92.16	91.41	84.47	92.39	93.32	87.74

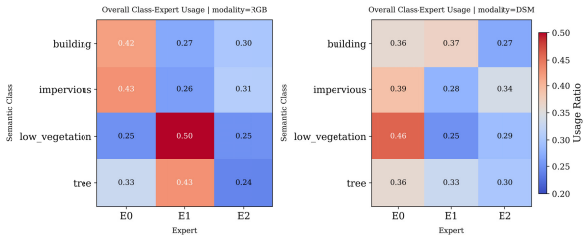


Fig. 8. Class-wise and modality-specific expert activation frequencies on the Vaihingen dataset.

TABLE VI

IMPACT OF THE NUMBER OF EXPERTS (E) AND THE NUMBER OF ACTIVATED EXPERTS (k) ON THE VAIHINGEN AND POTSDAM DATASETS (%)

Experts	Top-k	Vaihingen			Potsdam		
		OA	mF1	mIoU	OA	mF1	mIoU
$E = 3$	1	92.16	**91.41**	**84.47**	92.39	**93.32**	**87.74**
	2	91.66	91.01	83.78	92.20	93.03	87.25
$E = 4$	1	91.90	91.08	83.96	92.42	93.26	87.63
	2	91.33	90.04	82.25	**92.44**	93.26	87.64
	3	**92.34**	91.20	84.11	91.19	93.13	87.41

of 0.69% on Vaihingen and 0.49% on Potsdam over Top-2, and this trend remains consistent when $E = 4$. Although Top-3 performs best when $E = 4$ on Vaihingen, it still remains inferior to Top-1 when $E = 3$, confirming the advantage of sparse expert activation. Top-1 routing provides a sparse and decisive expert selection for each sample, which promotes clearer expert specialization. In contrast, activating multiple experts may weaken expert specialization and introduce noise from less relevant responses, thereby degrading segmentation performance. Notably, increasing the number of experts does not consistently improve performance. With Top-1 routing, increasing the expert number from $E = 3$ to $E = 4$ causes a 0.51% mIoU drop on Vaihingen and a marginal 0.11% decrease on Potsdam. Therefore, we adopt the optimal configuration in the subsequent experiments.

4) Effectiveness of Load Balancing Loss: The gating network tends to assign large weights to only a few experts, resulting in expert collapse. To alleviate this imbalance, we introduce a load balancing loss to encourage more balanced routing. In this subsection, we analyze the impact of the proposed $\mathcal{L}_{\text{lb}}$.

TABLE VII

IMPACTS OF LOAD BALANCING LOSS WEIGHT λ_{lb} ON THE VAIHINGEN DATASET (%)

λ_{lb}	OA	mF1	mIoU
0	91.74	91.00	83.77
0.005	92.13	91.00	83.76
0.01	**92.16**	**91.41**	**84.47**
0.02	92.10	91.18	84.06
0.05	91.74	90.57	83.09

TABLE VIII

PERFORMANCE COMPARISON OF INSERTING THE MOE ADAPTER AT DIFFERENT LAYERS ON THE VAIHINGEN DATASET (%)

layer	OA	mF1	mIoU
12	90.97	89.98	82.11
6,12,18	91.12	90.16	82.45
3,7,11,15,19,23	91.61	90.43	82.84
12,14,16,18,20,22	91.68	90.84	83.55
all layers	**92.16**	**91.41**	**84.47**

TABLE IX

COMPARISON RESULTS IN THE CONDITION OF FEW TRAINING SAMPLES ON THE POTSDAM DATASET. THE EVALUATION METRIC IS MIOU (%)

Method	2.5%	5%	10%
FTransUNet [17]	76.56	76.67	78.87
MFNet [29]	81.54	81.21	81.82
MultiMoE(Ours)	**83.70**	**85.32**	**84.45**

As shown in Table VII, introducing the load balancing loss improves segmentation performance. Compared with $\lambda_{lb} = 0$, using $\lambda_{lb} = 0.01$ increases the mIoU by 0.70%. However, further increasing λ_{lb} leads to performance degradation, indicating that excessive regularization is not beneficial. As a result, a small λ_{lb} is insufficient to regularize the routing behavior, while a large λ_{lb} may overly constrain the router and affect the main task. Therefore, we set $\lambda_{lb} = 0.01$ in the subsequent experiments.

5) Ablation Study of MoE Adapter Layers: To explore the best distribution of MoE adapters across ViT blocks, we conduct five experiments with different insertion layer configurations in the DINOv3 backbone. The specific layer settings and segmentation results are listed in Table VIII. As shown in Table VIII, the performance improves steadily as more layers are equipped with MoE adapters. In particular, inserting MoE adapters into all layers yields a notable mIoU gain of 2.36% compared with inserting only layer 12, indicating that multimodal interactions across more layers lead to stronger feature representations. In addition, placing MoE adapters in deeper layers achieves better results than using relatively shallow and sparse layers. Overall, the best performance is obtained when MoE adapters are inserted into all layers.

6) Performance in the Condition of Few Training Samples: To validate the effectiveness of the proposed MultiMoE and reveal its advantage under limited training data, we

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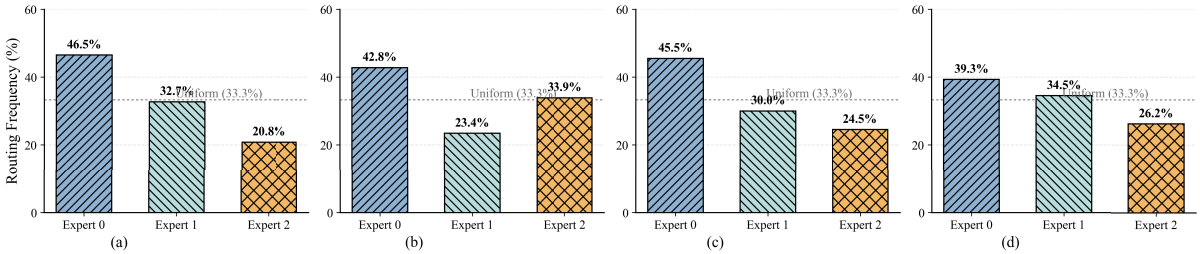


Fig. 9. Expert routing frequency on (a) Vaihingen-RGB, (b) Vaihingen-DSM, (c) Potsdam-RGB, and (d) Potsdam-DSM datasets. The statistics are computed over all inserted MoE layers. The dashed line indicates the uniform routing frequency of 33.3% for three experts.

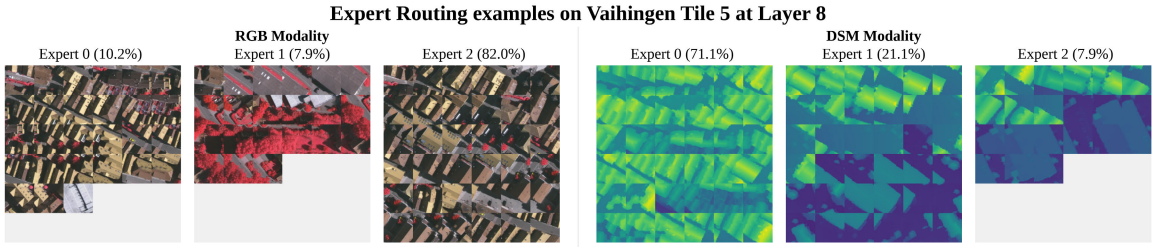


Fig. 10. Sample-level expert routing examples on Vaihingen tile 5. The left and right groups show the routing patterns of RGB and DSM samples, respectively, from Vaihingen tile 5 at the MoE layer inserted in the eighth encoder block.

conducted a comparative experiment with different proportions of the Potsdam dataset. Specifically, 2.5%, 5%, and 10% of the training samples are selected for model training, while the entire test set is used for evaluation. The results are reported in Table IX. MultiMoE consistently achieves the best performance under all three settings. In particular, compared with FTransUNet and MFNet, MultiMoE improves the segmentation accuracy by 7.14% and 2.16%, respectively, when only 2.5% of the training samples are used. This advantage is still maintained under the 5% and 10% settings. The promising performance of MultiMoE under limited training samples benefits from the combination of the frozen foundation model and MoE adapters. DINOv3 provides strong visual priors learned from large-scale pretraining, reducing the dependence on abundant task-specific annotations. Meanwhile, the lightweight MoE adapters serve as task-specific adaptation modules for the frozen backbone. Their dynamic routing mechanism selects suitable experts for each sample, enabling flexible adaptation to new remote sensing scenes with limited trainable parameters.

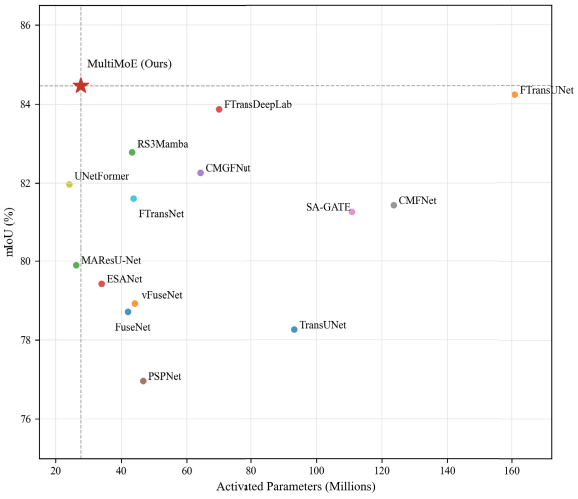


Fig. 11. Comparison of MultiMoE (ours) with representative semantic segmentation models in terms of activated parameters and mIoU. The x-axis represents the number of activated parameters (millions), while the y-axis indicates the segmentation accuracy measured by mIoU. (%).

V. Model Scale Analysis

The improved performance of MultiMoE is largely attributable to the strong backbone of the VFM DINOv3. As a large-scale model, DINOv3 provides rich generalization ability. However, it does not exhibit advantages in computational complexity or inference speed compared to existing lightweight models. Consequently, we focus on reporting the number of trainable parameters and memory usage to evaluate the hardware requirements.

Table X summarizes the model scale statistics of all competing methods. The proposed multimodal framework with MoE enables large VFMs to operate on a single GPU while maintaining manageable trainable parameters and memory consumption. Owing to the sparse activation mechanism, MultiMoE can increase the total model capacity without introducing expensive computational cost. Specifically, although the total number of parameters reaches 78.17 M, only 27.67 M parameters are activated during inference. As a result, the effective number of active parameters in MultiMoE is comparable to the total parameters of lightweight models such as UNetFormer and MAResU-Net, and remains significantly lower than that of high-capacity models such as CMFNet and FTransUNet. As illustrated in Fig. 11, MultiMoE achieves superior parameter efficiency while maintaining competitive

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TABLE X

MODEL SCALE ANALYSIS MEASURED ON A 256 × 256 IMAGE USING A SINGLE NVIDIA GEFORCE RTX 4090 GPU. THE REPORTED PARAMETERS CORRESPOND TO THE TRAINABLE PARAMETERS OF THE DINOV3-BASED MODEL, INCLUDING THE MOE MODULES AND DECODER. THE NUMBER IN PARENTHESES DENOTES THE ACTIVATED PARAMETERS PER SAMPLE UNDER SPARSE MOE ROUTING. MIOU VALUES ARE EVALUATED ON THE VAIHINGEN DATASET. THE BEST RESULTS ARE IN BOLD

Method	Parameter (M)	Memory (MB)	mIoU (%)
FuseNet [16]	42.08	2284	78.71
vFuseNet [45]	44.17	2618	78.92
MAResU-Net [46]	26.27	1908	79.89
ESANet [47]	34.03	1914	79.42
CMGFNet [48]	64.20	2463	82.26
PSPNet [49]	46.72	3124	76.96
SA-GATE [50]	110.85	3174	81.27
CMFNet [8]	123.63	4058	81.44
UNetFormer [41]	**24.20**	1980	81.97
FTransNet [9]	43.77	1549	81.61
TransUNet [51]	93.23	3028	78.26
FTransUNet [17]	160.88	3463	84.23
RS3Mamba [52]	43.32	**1548**	82.78
FTransDeepLab [53]	69.86	1624	83.87
MultiMoE	78.17 (27.67 active)	3152	**84.47**

segmentation performance. Despite the reduced computational footprint, it attains 84.47% mIoU and outperforms most competing methods.

VI. CONCLUSION

This article proposes a MultiMoEs fine-tuning framework for remote sensing image semantic segmentation. By integrating lightweight MoEs adapters into the DINOv3 backbone, the proposed framework can capture both general visual knowledge and modal-specific contexts effectively. To encourage expert specialization across modalities, a shared gating network combined with modality-specific expert pools was designed to dynamically choose experts while preserving the unique characteristics of each modality. Extensive experiments conducted on two widely used datasets demonstrated that our MultiMoE achieved superior performance compared to current state-of-the-art segmentation methods. In the future, we will continue to explore the promising and effective parameter-efficient learning paradigm for multimodal remote sensing image semantic segmentation.

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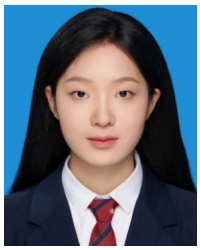
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